

Michael Farquharson

905-616-8182 | mfarquharson1064@gmail.com | <https://www.michael-farquharson.com/>

EDUCATION

McMaster University

BEng in Mechanical Engineering and Management

Hamilton, ON

Sept. 2015 – Apr. 2021

- GPA 3.6/4.0 (Summa Cum Laude); entrepreneurship-focused engineering and commerce curriculum.

McGill University

MSc in Mechanical Engineering

Montreal, QC

Sept. 2021 – Aug. 2025

- Research in teleoperation, contact-rich rigid-body simulation, and real-time model predictive control in partnership with NRC-AMTC.

EXPERIENCE

HARP Technical Lead / Robotics Software Specialist

Dec. 2023 – Present

Haply Robotics

Montreal, QC

- Led development of the HARP teleoperation platform, including a Rust-based concurrent control interface for haptic devices.
- Architected low-latency control and modular safety features to support robust, scalable deployments.
- System designed to provide an intuitive solution to collecting high quality data for Physical AI training.
- Managed a cross-functional team and supported demos, customer feedback, and technical stakeholder reviews.

Graduate Student Researcher

Sept. 2021 – Aug. 2025

McGill University

Montreal, QC

- Built teleoperation simulations with contact physics for remote aerial vehicles.
- Developed ROS 2-based test platforms for low-latency model predictive control experiments.
- Designed and evaluated haptic control algorithms for human-machine interfaces.

Data Analyst Intern

May 2019 – Oct. 2019

Bombardier

Toronto, ON

- Automated data cleaning and analysis workflows with Python to drastically reduce recurring task time.
- Delivered quarterly presentations to senior management highlighting efficiency gains.

PROJECTS

HARP Teleoperation Control Platform | *Rust, Concurrency, Haptics*

2025 – Present

- Designed a low-latency robot control platform for force-feedback teleoperation devices.

Teleoperation Research Platform | *ROS 2, Simulation, Haptics*

2022 – 2025

- Built contact-rich rigid-body simulations and a ROS 2 testbed to study haptic teleoperation.

Custom Mechatronic Prototyping | *3D Printing, Control, Hardware*

2023 – 2024

- Rapidly prototyped human machine interface devices and iterated on mechanical and control system designs.

TECHNICAL SKILLS

Languages: Rust, Python, C/C++, MATLAB, Bash

Robotics & Control: Teleoperation, Haptics, ROS 2, Control Systems, Real-time Simulation, Model Predictive Control

Tools: Git, Linux, SolidWorks, Onshape, Autodesk Inventor, Tauri, MuJoCo

Prototyping: 3D Printing, Soldering, Rapid Prototyping, Mechatronic Integration